

Olive Oil: Extraction Technology, Chemical Composition, and Enrichment Using Natural Additives

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Abstract

Virgin Olive oil (VOO) is considered the primary source of added fat in the Mediterranean diet. Its consumption is linked to numerous health-promoting properties along with its high energetic value. These properties are the results of various chemical compounds (fatty acids, tocopherols, polyphenols, etc.). VOO provides monounsaturated fatty acids, which lower total cholesterol and low-density lipoprotein cholesterol levels. VOO is obtained by three mechanical extraction processes, which can be classified into two systems that can be followed to extract olive oil from olives: the so-called traditional or discontinuous method, and the modern or continuous one. After the extraction of olive oil, its oxidative stability and chemical composition are subjected to deterioration especially when stored under inappropriate conditions (light, O₂, temperature, etc.). To deal with the problem, VOO enrichment using natural additives became an important practice to enhance VOO oxidative stability and its chemical composition. In this chapter, various aspects related to VOO extraction processes, chemical composition, stability oxidative and enrichment via natural additives will be reviewed and discussed in light of published literature.

Keywords: natural additives, chemical composition, enrichment, extraction technologies, olive oil

1. Introduction

Olive oil is one of a great interest in the vegetable oils world market. It is produced from the fruit of olive (*Olea europaea* L.). Virgin olive oil (VOO) is obtained exclusively by mechanical cold extraction [1]. It is not subjected to any chemical treatments apart from washing, decantation, centrifugation, and filtration. These processes may be carried out without refining, which makes the obtained oils highly appreciated by consumers thanks to their rich nutritional value, several health benefits, and unique organoleptic properties. Olive oil organoleptic and nutritional characteristics arise from noble compounds it contains. VOO composition consists of an unsaponifiable fraction (1 to 2%) along with essential unsaturated fatty acids contained in glyceridic

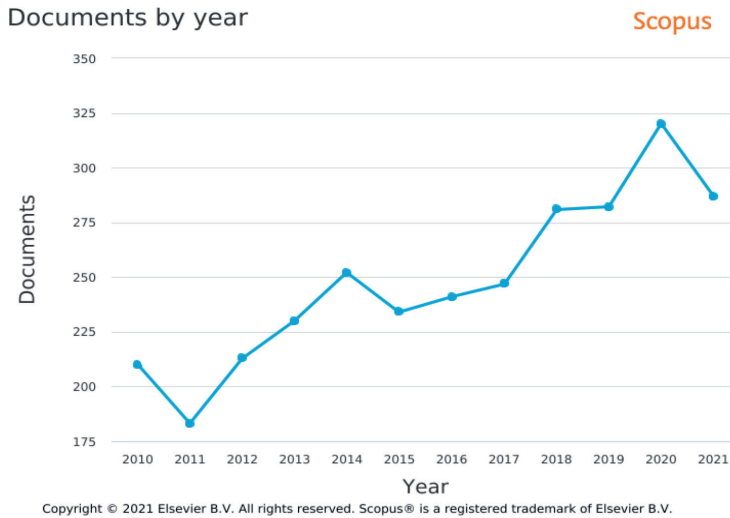


Figure 1.

Publication trends of olive oil composition (based on data retrieved from Scopus database).

fraction (98 to 99%) [2]. The composition of olive oil is well outlined in the literature. An updated analysis of the composition of olive oil reported in the literature is shown in **Figure 1**. In fact, on November 29th, 2021, the Scopus database was chosen to search for peer-reviewed literature regarding olive oil composition. The search string: (“olive oil*” AND “composition*”) was utilized to extract bibliometric information from the Scopus online database. A total of 2980 publications were recovered through the literature search within the range of years from 2010 to 2021, 1892 of them representing about 65% are published in the interval of years from 2015 to 2021 (**Figure 1**).

2. Olive oil extraction technologies

Olive oil is made from fresh olives, which are extracted by mechanical processes [3]. Olive oil extraction technologies are summarized in **Figure 2**. There are two main olive oil extraction processes: traditional oil mills, and a relatively new extraction process known also by continuous mills and characterized by two or three phases [4, 5] **Figure 2**. All the above processes aim at separating the liquid oil phase from the other constituents of the fruit [6]. Likewise, olives should be processed as rapidly as possible after harvesting to reduce oxidation and preserve their quality [7].

Concerning the traditional press method, olive fruits liberated from leaves are washed, crushed using mill stones, and malaxed into a paste containing solid matter (core debris, epidermis, cell walls, etc.) and fluids (oil and vegetation water contained in the cells of olives). This is then spread on spherical mats [6]. Pressure with a hydraulic piston press is exerted then to obtain, firstly, a solid fraction (known as pomace) and, secondly, the mixture of oil and water is filled into a container and, eventually, the oil and water are then separated by gravity and collected through decantation [5]. The pressing process is the oldest method of obtaining olive oil [8]. Owing to lower production efficiency and high labor costs, during the last decade, the discontinuous pressing systems have widely been substituted by continuous systems, along with the development of centrifuge technology [4, 9]. After the steps of washing, crushing, and

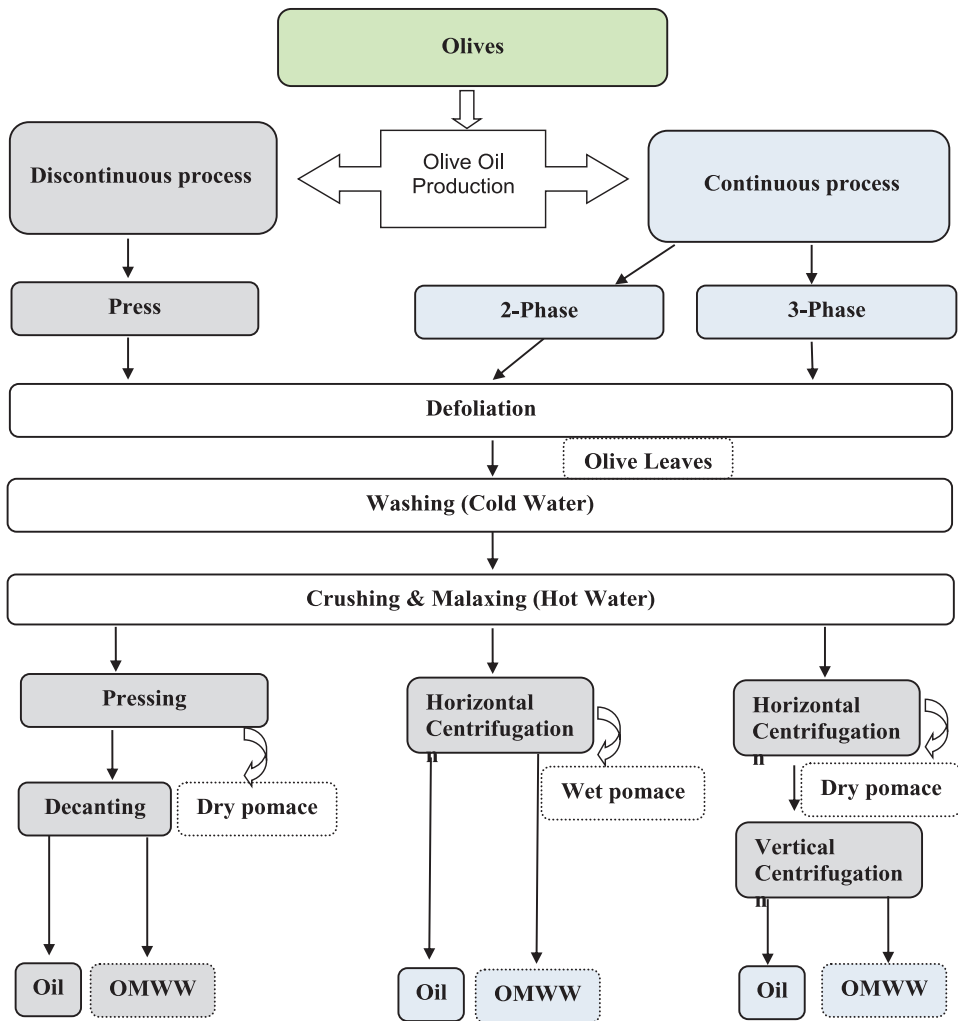


Figure 2.
Scheme of discontinuous and continuous extraction systems. OMWW = olive mill wastewater.

mixing, the mechanical extraction of the oil occurs mainly by a continuous process based on centrifugation using a decanter. The decanter centrifuge is equipped with a rotary bowl as well as a screw conveyor, which allows the processing of great quantities of olives in a short time [7]. Continuous separation systems can be divided into two-phase and three-phase systems, based on the decanter type used and the level of the phase of separation [9]. In the three-phase process, an additional amount of hot water is added to wash the oil, and then the three-phase decanter (insoluble solids, oil phase, and an aqueous phase), are separated following their density [7, 10]. Firstly, the solid wastes (insoluble solids), are separated from the remaining two phases in the decanter, and the liquid phases (oil phase as well as aqueous phase), are then subjected to vertical centrifugation to separate the olive oil from the olive mill wastewater [7].

Owing to the significant issue of wastewater produced, this three-phase system is preferred over the two-phase system since it is more eco-friendly [11]. This latest uses only a semi-liquid slurry (vegetation water along with insoluble solids) phase and the

oil phase, a semi-liquid slurry, which is also known as two-phase olive mill waste [7]. This process has a reduced environmental impact owing to the reduced requirement of water as well as the amount of waste produced [7].

3. Olive oil composition

3.1 Bioactive compounds

Olive oil glyceridic fraction consists of triacylglycerols, diacylglycerols, monoacylglycerols and free fatty acids (FFA). Among them, 80% of them are unsaturated fatty acids. It is particularly rich in essential monounsaturated fatty acids (55–83% of oleic acid) and polyunsaturated fatty acids (2.5–21% of linoleic acid) [12]. The remaining fatty acids, apart from C16: 1, display an average value ranging from 0.3 to 3.5% (**Table 1** and **Figure 3**). Nevertheless, linolenic acid is a minority and its concentration is lower than 1% [12]. A low level of linolenic acid can be used to detect adulteration via some vegetable oils such as rapeseed and soybean oils [13]. Small quantities of saturated fatty acids also compose the triglycerides of olive oil: stearic acid (about 0.5–5%) and palmitic acid (about 7.5–20%). The remaining fatty acids (C17: 0, C17: 1, C20: 0, C20: 1, and C22: 0) are found to be of lower magnitudes. Since their concentrations are below 0.5% (**Table 1**). The unsaponified matter (about 1–2%) contains sterols, triterpene alcohols, tocopherol (mainly α -tocopherol), tocotrienol polyphenols, and squalene. The oil also contains a non-negligible proportion of volatile compounds. The total phytosterols content of VOO ranges between 100 and 200 mg/100 g. Also, 100 mg/100 g represents the inferior

Fatty acid [g/100 g]	Norm [12]	Physicochemical parameters	
Myristic acid [C14: 0]	≤ 0.2	Density [20°C]	0.906–0.919
Palmitic acid [C16: 0]	11.5–15	Refraction index [20°C]	1.463–1.472
Stearic acid [C18: 0]	4.3–7.2	Saponification value [mg of KOH/g]	184–196
Arachidic acid [C20: 0]	≤ 0.5	Iodine value [g (I ₂)/100 g]	75 to 94
Behenic acid [C22: 0]	≤ 0.2	Phytosterol [g/100 g]	(IOC 2021)
Σ SFA	15.8–23.1	Cholesterol	≤ 0.5
Palmitoleic acid [C16: 1]	≤ 0.2	Brassicasterol	≤ 0.1
Oleic acid [C18: 1]	43.0–49.1	Campesterol	≤ 4
Eicosenoic acid [C20: 1]	≤ 0.5	Stigmasterol	< Campesterol
Σ MUFA	43–49.8	Delta-7-stigmastanol	≤ 0.5
Linoleic acid [C18: 2]	29.3–36.0	Apparent beta-sitosterol	> 93
Linolenic acid [C18: 3]	≤ 0.3	Total sterol [mg/100 g]	≤ 220
Σ PUFA	29–36.3	Erythrodiol & Uvaol (% total sterols)	≤ 4.5

SFA-Saturated Fatty acids, MUFA-Monounsaturated fatty acids, PUFA-Polyunsaturated fatty acids: Apparent beta-sitosterol: beta-sitosterol + delta-5-avenasterol + delta-5-23-stigmastadienol + cleroesterol + sitostanol + delta 5-24-stigmastadienol.

Table 1.
Physicochemical parameters, fatty acids, phytosterols, and tocopherols composition of olive oil.

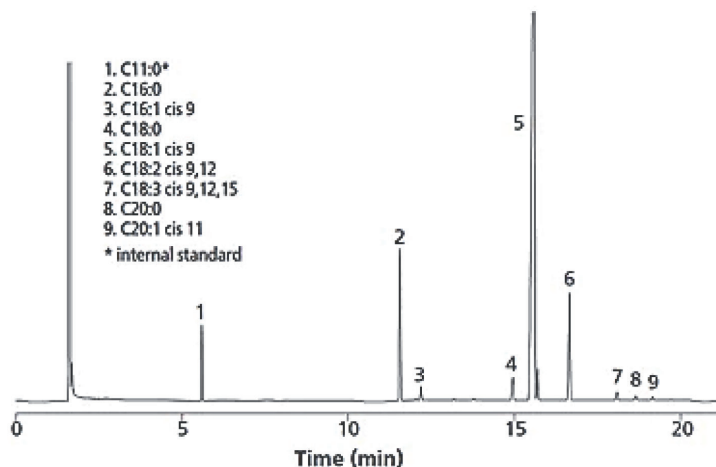


Figure 3.
 Chromatogram of fatty acids.

limit set by the international olive council [12]. Apparent beta-sitosterol, (beta-sitosterol + delta-5-avenasterol + delta-5-23-stigmastadienol + clerosterol + sitostanol + delta 5-24-stigmastadienol) are the main compounds in the sterol fraction with a value more than 93% while β -sitosterol has the greatest relative percentage [14, 15] (**Figure 4**). VOO content also includes up to 4.5 g/100 g of total phytosterols [12]. The erythrodiol (5 α -olean-12-ene-3 β , 28-diol, homo-olestranol) in free and esterified forms and are the major triterpene di-alcohols found in olive oil [14], and their percentage reached up to 4.5% of the total content of sterols [12]. Moreover, four isoforms of tocopherols (α , β , γ , and δ -tocopherol) (**Figure 5**) and four tocotrienols (α , β , γ and δ -Tocotrienol) are present in olive oil. α -tocopherol is the main tocopherol found in olive oil, constituting more than 90% of the total tocopherol fraction [14]. Cunha et al. [16] reported that the proportions of tocopherols and tocotrienols ranged from 100 to 270 mg/kg in Portuguese olive oils [16]. Gharby et al. [47] found that the values of tocopherols varied from 150 to 250 mg/kg in three varieties ('Arbequina', 'Moroccan Picholine', and 'Picual') of olive oil [17]. Moreover, another study, based on the comparison of the tocopherol contents of olive oils from 4 different varieties harvested at different ripening periods found

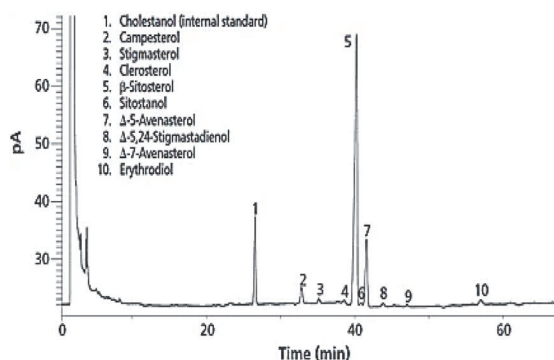


Figure 4.
 Chromatogram of sterols.

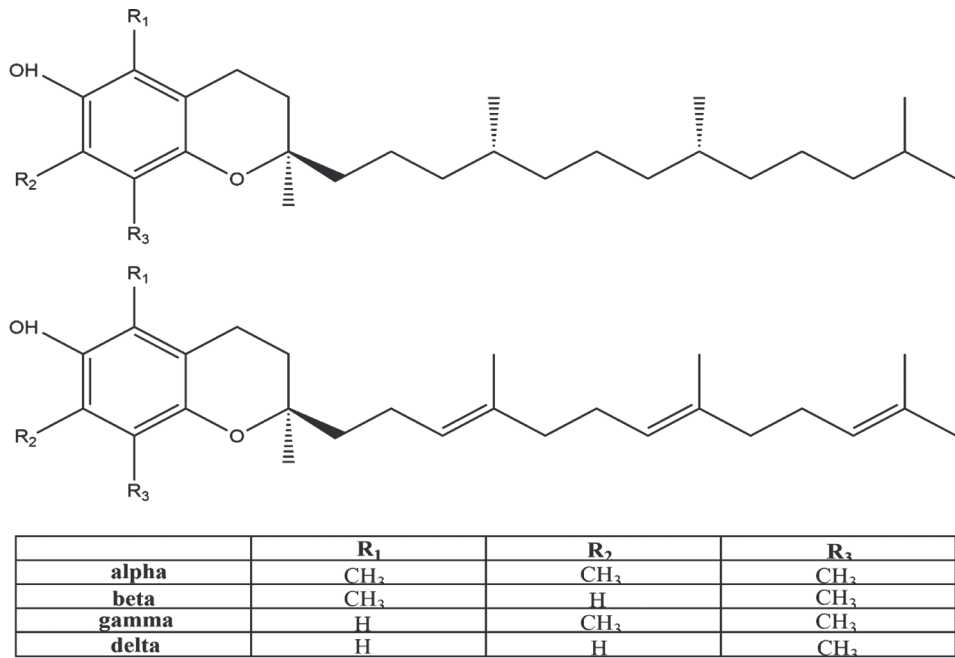


Figure 5.
Tocopherols chemical structure.

that the α -tocopherol (major tocopherol) in oils obtained from olives composed of 130.54–180.43 mg/kg [18]. In general, tocopherol and tocotrienol levels in oil fluctuate with several factors such as harvest year, climatic conditions, storage time, extraction method, soil properties and spacing between olive trees [19]. Tocopherols possess a strong antioxidant power [20]. Together with tocopherols and tocotrienols, olive oil contains other antioxidant molecules such as polyphenolic compounds.

Many research works have demonstrated that the content of tocopherols in VOO is lower than that of argan oil [21–23].

The phenolic compounds are endowed to have a large scale of biological functions including stability to auto-oxidation, beneficial effects on human health [24]. About their well-known activities, olive oil polyphenols have been proven to possess an effective role in maintaining the organoleptic properties and the stability of olive oils [25].

Such bioactive compounds are extensively studied for their anti-inflammatory, antioxidant, neuroprotective, cardioprotective, antidiabetic, antimicrobial, and anticancer properties [26–29].

Franco et al. reported that phenolic compounds have a considerable increase during olive fruit growth. However, they are reduced when the fruits reach the maturation stage [30]. Khalatbary documented that the total phenolic content (TPC) in olive oils varies from 190 to 500 mg/kg [31]. In addition, in extra virgin olive oil, TPC commonly varies from 250 to 925 mg/kg [32]. Other factors including climatic conditions, variety, storage time, extraction conditions, soil properties, and analysis of polyphenolic compounds can lead to important variations in TPC [33]. Likewise, several classes of polyphenols are found in olive oils. These are presented as a separate class, to better understand the antioxidant phenolic chemistry of olive oil [33]. Finicelli et al. classify olive oil polyphenols following their chemical structure as follows [34]:

- Phenolic alcohols with a hydroxyl group are linked to an aromatic hydrocarbon group. The main constituents of this class are oleocanthal, hydroxytyrosol, and tyrosol (**Figure 6**) [26].
- Secoiridoids are phenolic compounds present in high amounts in olive oil in comparison to other plant species. The bitterness of extra virgin olive oil is a result of the content of secoiridoids [35].
- Lignans are chemically characterized by the aggregation of aromatic aldehydes. The pulp of the olives as well as the woody part of the seed contains lignans. These molecules are liberated into the oil during the process of extraction without biochemical changes [36].
- Flavonoids are chemically structured with two benzene rings attached via three linear carbon chains. The first flavonoids identified in VOO were flavones; their free forms, apigenin, and luteolin. They are the more abundant compounds [36].
- Hydroxyisocromans are the only two molecules characterized in commercial VOO. These compounds are produced via the HydroxyTyrosol reaction with benzaldehyde and vanillin [37].
- Phenolic acids are divided into two main classes: hydroxycinnamic acid along hydroxybenzoic acid [26].

The volatile fraction of VOOs has been reported to have about 280 different compounds [38]. The majority of volatile compounds are quickly developed during olive milling as a result of the disturbance of olive cells [39]. Although, Nardella et al. reported that most of the volatile compounds typical of olive oils are generated during

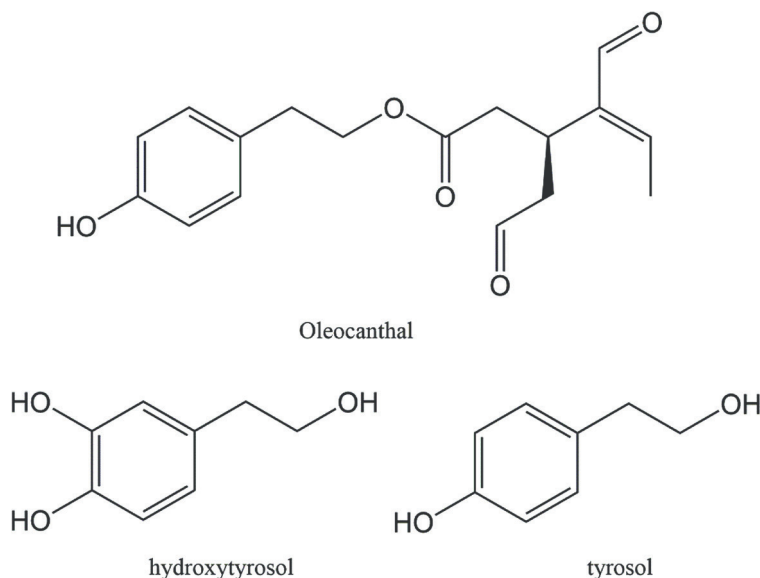


Figure 6.
 Some phenolic alcohols present in olive oil.

malaxation due to the activation of particular pathways, in which the lipoxygenase (LOX) enzyme plays an essential role in producing a large quantity of C6 aldehydes, esters, and alcohols. These constitute almost all of the positive sensory marks in olive oils [40]. Such changes are initiated when olive tissues are affected, thereby enhancing the liberation of endogenous enzymes like hydroperoxide lyase and lipoxygenase [40].

Besides, several analytical techniques have been used to determine volatiles composition in olive oil. The main important are: GC (gas chromatography), HPLC (high-performance liquid chromatography), HPLC/MS (high-performance liquid chromatography/mass spectroscopy), IRMS (isotope ratio mass spectroscopy), ICP (inductively coupled plasma spectroscopy), NMR (nuclear magnetic resonance), SPME-GC/MS (solid-phase microextraction followed by gas chromatography/mass spectrometry), SNIF/NMR (specific natural isotopic fractionation nuclear magnetic resonance), SCIRA (stable carbon isotope ratio analysis), PTR/MS (proton transfer mass spectrometry) [41]. Fregapane et al. reported that the composition of volatiles may be affected significantly according to many factors such as cropping season, olive variety, harvest time, technological parameters, and agronomic conditions among other factors [42]. Ghanbari et al. reported that several chemical factors such as hydrophobicity, volatility, position, and functional groups type are reported to be directly linked to the odor degree of a given volatile component more than its content [38]. Theodosi et al. investigated correlations between the composition of volatiles of olive oil and altitude variation. The findings demonstrate that the total volatile compounds of 'Koroneiki' olive oil samples and altitude levels are negatively associated. The most important volatile compounds are alcohols, aldehydes, esters, and hydrocarbons [43].

3.2 Physicochemical parameters

Parameters routinely used to evaluate physicochemical properties of olive oil include density, iodine value, refractive index, saponification value along with unsaponifiable matter. For vegetable oils including olive oil, both density and refraction index depend on the temperature [44]. **Table 1** shows the ranges of the main physicochemical parameters of olive oil. The refractive index at 20°C varies in the range 1.463–1.472. At this same temperature, its density relative to water is between 0.906 and 0.919 [12]. The iodine value is a measure of the total number of double bonds found in an oil sample [1]. Olive oil displays an iodine value between 75 and 94 mg/100 g [17] (**Table 1**). This value is lower than that of argan oil (91–110 g I₂/100 g), and cactus seed oil (131.5 ± 0.5 g (I₂)/100 g) but higher than that of coconut oil (6.3–10.6 g (I₂)/100 g) [13]. High iodine value is associated with the greater number of double bonds and reduced oxidative stability [45]. The saponification value is a measure of the average chain lengths of fatty acids. An oil sample with shorter fatty acids has a high saponification value. Moreover, according to CODEX STAN 33, the saponification value of olive oil varies between 184 and 196 mg (KOH)/g.

4. Quality control of olive oil

Olive oil is subject to enormous analytical and sensory controls to assess its overall quality. These analyses evaluate the freshness of the oil regarding hydrolytic and oxidative alterations to ensure the conformity of products to their labels. For example, extra VOO by simple routine analyses (free fatty acids, peroxide value, specific extinction (E270 along with E232) and/or purity blending with other oils and

contaminants. These criteria require detailed analyses (triglycerides contents, fatty acids, sterols, tocopherols, etc. ...). Organoleptic characteristics (taste, odor, color, etc. ...) also have to be taken into account.

As for other vegetable oils, the olive oil oxidation leads to natural phenomena alteration [46, 47]. This can be controlled since fruit harvest until oil storage. Because of oxidation, physicochemical parameters such as acidity, peroxide value and extinction specific at wavelength 270 (λ_{270} or λ_{270}) have been selected as the backbone of olive oil quality determination by the International Olive Council [12]. Also, acidity of olive oil is classified into four grades: extra-virgin (Acidity < 0.8 g/100 g), fine-virgin (0.8 < Acidity < 2 g/100 g), ordinary virgin (2 < acidity < 3.3 g/100 g), and lampante olive oil (Acidity > 3.3) (Table 2) [12].

The variability of the extra VOO, acid value according to various parameters has been studied [47]. Oil oxidative state is examined from peroxide value and specific extinction coefficients (K_{232} or K_{270}). These indicate the presence of primary and secondary oxidation products [1, 48]. The peroxide value of extra VOO oil must be below 20 mEq O_2 /kg and specific extinction K_{232} < 2.5. The other two main indices used to evaluate the secondary oxidation products are the following: p-anisidine value and specific extinction K_{270} [1, 49]. The International Olive Council (IOC) has set 0.22 and 0.25 as a limit value for both the extra VOO and VOO, respectively [12].

Furthermore, along with oxidation and acidity concerns, the quantification of major compounds such as fatty acids (Figure 3), and minor compounds, like sterols (Figure 4), polyphenols, tocopherols, minerals elements, and other bioactive molecules, are also of great importance for the purity and for detection of olive oil adulteration, which is a complex problem. Owing to its high cost and demand, fraudsters blend VOO with cheaper edible oils (most often with sunflower and soybean oils) and sometimes with low-quality olive oil. Today, the problem exceeds the borders of the main producer countries and it tackles the international level market. In addition to known risks of commercializing a mixture of vegetable oils. There is another type of adulteration resulting from the mixing of relatively low and high-quality olive oil, and the outcome is a product, which is sold as “high quality extra VOO”. The control of adulteration, and authentication is of a crucial importance for the olive oil quality control. Codex Alimentarius (fats and oils), International Olive Council, and European Union Commission are dealing with the monitoring along with the regulation of VOO [50]. These international organizations have described the official control methods and have specified olive oil quality limits. Generally, all analytical techniques (chromatography, spectrophotometry, voltametric, differential scanning calorimetry), as well as several analytical methods, have been used to detect the adulteration of olive oil. Gas

Category olive oil	Acidity (g/100 g)	Peroxide index mEq O_2 /Kg	Extinction specific at K_{232}	Extinction specific at K_{270}
1. Extra virgin olive oil	≤ 0.8	≤ 20	≤ 2.50	≤ 0.22
2. Fine Virgin olive oil	≤ 2.0	≤ 20	≤ 2.60	≤ 0.25
3. Ordinary Virgin olive oil	≤ 3.3	≤ 20	no limit	≤ 0.30
4. Lampante olive oil	> 3.3	no limit	no limit	no limit

Table 2.
Limits established for acidity, peroxide index and extinction specific (K_{232} and K_{270}) for each olive oil category.

chromatography (GC), which analyzes oil fatty acids profile, can be used to detect virgin oil purity by distinguishing it from other vegetable oils such as sunflower, soybean, walnut, rapeseed, and canola oils [51]. Moreover, HPLC-technique can be used, to calculate, the difference between the theoretical and experimental equivalent carbon number ($\Delta EC_{N42_{th}}$). Likewise, the determination of phytosterols composition (namely campesterol $\Delta 7$ -stigmasterol) using gas chromatography can be used to detect olive oil adulteration with low levels of cotton, corn, sunflower, soybean, and rapeseed oils [51]. In addition, Vietina et al. reported that the polymerase chain reaction (PCR) technique was demonstrated to be an efficient technique to detect VOO adulteration with cheaper vegetable oils by comparing their DNA melting profiles [52]. MS has also been used to detect the fraudulent presence of vegetable oils. Also, a lot of different techniques involving MS have been significantly developed, such as LC-MS, GC-MS, and MALDI-TOF/MS, which are of highly accurate identification [51]. Indeed, many other studies have also outlined the application of fluorescence spectroscopy, UV-Vis spectroscopy, [50] Fourier transform infrared spectroscopy [53] mid-infrared (MID) or near-infrared spectroscopy (NIR) [54] and Raman spectroscopy [55] for authentication and detection of adulteration of vegetables oil present in VOO [50]. Otherwise, differential scanning calorimetry (DSC) has also been used to detect argan oil purity by discriminating it from sunflower, high oleic sunflower as well as refined hazelnut oil [50]. Apetrei and Apetrei have investigated the use of the voltametric method based on modified EO

Negative attributes		
Fusty/muddy sediment	The characteristic flavor of oil obtained from olives stocked in such a way that they have developed an enhanced stage of anaerobic fermentation,	
Musty-humid-earthly	Flavor characteristic of oils from fruits in which a lot of fungi and yeasts have been developed or from olives picked up with mud or earth and not been previously cleaned.	
Winey- vinegary	The flavor characteristic of some oils reminiscent of vinegar or wine.	
Acid-sour	The flavor is primarily caused by the formation of ethyl acetate, ethanol, and acetic acid.	
Rancid	Flavor of oils that have been submitted to an intense oxidation process	
Frostbitten olives (wet wood)	Flavor characteristic of oils obtained from olives that have been damaged by frost on the tree.	
Positive attributes		
Fruity	Characteristic of the oil that varies according to the variety and is obtained from fresh olives, ripe or not.	
Bitter	Primary taste characteristic of oil extracted from green olives or olives that are becoming colored.	
Pungent	Characteristic of oils obtained at the beginning of the crop year, mostly from olives that are not ripe yet.	
	Median of defect (Md)	Fruity median (Mf)
1. Extra virgin oil	Md = 0	Mf > 0
2. Virgin olive oil	Md ≤ 3.5	Mf > 0
3. Lampante oil	Md > 3.5	—

Table 3.
Organoleptic attributes of olive oil.

carbon paste-based sensors to determine the adulteration of VOO with soybean and sunflower oils [56].

On the other hand, identification of contaminants is one of the multiple checks that must be performed on oils. Vegetable oils have limited values for aromatic hydrocarbons polycyclics (PAHs), heavy metals, mycotoxins, phthalates, and pesticides. Although, the physicochemical characterization of olive oil is an essential step, it is not sufficient and organoleptic characteristics along with the above-mentioned supplementary analyses are required for a full picture of olive oil quality [1].

To satisfy consumers, organoleptic characteristics (color, taste, smell, etc.) must be taken into account. This is particularly important for olive oil. The organoleptic analysis is an essential step for successful food marketing. It is an integral part of evaluating olive oil. IOC has established a procedure to evaluate the organoleptic characteristics of VOO according to COI/T.20/Doc. [12] No 15/Rev. 102,018. It has classified such characteristics into positive and negative attributes as highlighted in **Tables 2** and **3**.

5. Olive oil enrichment with natural additives

Oxidation of lipids including oils is a major concern to food industries [57, 58]. While, vegetable oils are endowed with a wide variety of endogenous antioxidants (pigments, vitamins, tocopherols, phenols, etc.), the use of exogenous antioxidants is widely practiced to enhance oxidative stability. In this regard, synthetic antioxidants such as butylated hydroxyanisole (BHA), butylated hydroxytoluene (BHT), tertiary butylhydroquinone (TBHQ), as well as propyl gallate are commercially used to extend oils' shelf life by delaying or even hindering lipids degradation. These molecules are considered as Generally Recognized as Safe (GRAS) preservatives with a concentration limit of 0.02% in oils and fats [59]. In contrast, some reports associated these molecules with health risks because of carcinogenesis, leading to a restriction of the use of the GRAS list and a reduction of their utilization in different countries [59]. For this reason, natural antioxidants are a good alternative to replace the synthetic ones in preserving vegetable oils including olive oil [59–61]. An overview of factors involved in the balance of antioxidants and pro-oxidants as well as synthetic and natural antioxidants are summarized in **Figure 7**.

Natural extracts sourced from various plant parts (peel, fruit, leaf, flower, and root) from different aromatic and medicinal herbs, agri-food residues and by-products were investigated for their antioxidant power as well as their use for the enrichment of olive oil with an emphasis on improving oxidative stability. Such natural extracts were proved to have a wide range of bioactive compounds were identified. These are mainly carotenoids and phenols [62, 63]. Promising results were obtained regarding the improvement of oxidative stability and shelf life of olive oil. Regarding the antioxidant activity of synthetic and natural additives, several mechanisms are involved. They act as free radical scavengers, inactivators of peroxides as well as other reactive oxygen species (ROS), singlet oxygen quenchers, metal ion chelators, quenchers of secondary oxidation products, and inhibitors of pro-oxidative enzymes, among other compounds [64]. Following these authors, antioxidants can be classified, based on their mode of action, into primary antioxidants. These break the oxidation chain reaction through scavenging free radical intermediates, however secondary antioxidants delay or even prevent oxidation through suppression of oxidation initiator, accelerators or regeneration of primary antioxidants.

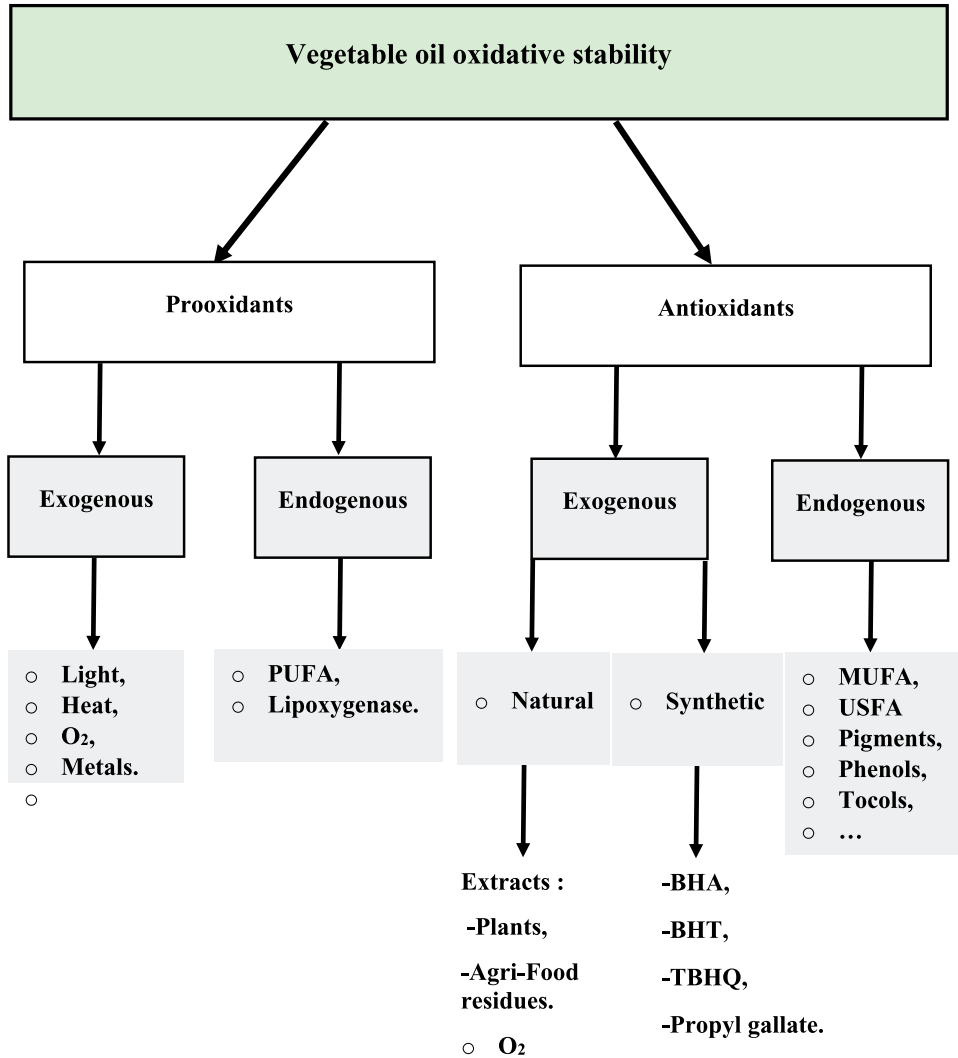


Figure 7.
An overview of factors involved in olive oil oxidative stability as well as natural and synthetic antioxidants. BHA, butylated hydroxyanisole; BHT, butylated hydroxytoluene; TBHQ, tertiary butylhydroquinone; MUFA, monounsaturated fatty acids, and PUFA, polyunsaturated fatty acids.

6. Conclusions

Olive oil is an important food in the Mediterranean diet. Its importance and nutritional value arise from chemical composition. Its richness in essential fatty acids is behind their health-promoting properties. A set of other minor compounds such as polyphenols and tocopherols act as antioxidants which are directly associated with oxidative stability and shelf life of olive oil on one hand as well as human health on the other hand. Along with these endogenous antioxidants, olive oil quality can be enhanced through natural antioxidants extracted from herbs and agri-food residues.

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Conflict of interest

The authors declare no conflict of interest.

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